



Semiannual Progress Report

2026-07

Student

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Admission Cohort	2024-09	Months of Study	18

I. Summary of the progress of your research work and existing problems since last progress report

Please attach a short technical report on your research work.

Since the last progress report, my research has focused on a new pan-cancer single-cell project investigating heat shock protein (HSP)-associated cellular states in the human tumor microenvironment. I curated human HSP genes from HGNC and established a reproducible analysis workflow that processes each lineage and subtype separately, including reclustering, HSP gene expression profiling, HSP score visualization, manual HSP-high cluster definition, differential expression, GO enrichment, trajectory analysis, alternative stress-program scoring, and within-subtype CellChat analysis where a reliable HSP-high cluster is present.

The current analysis covers 57 cell subtypes across lymphoid cells (B, CD4 T, CD8 T, and NK cells), myeloid cells (macrophages and dendritic cells), and stromal cells (fibroblasts and endothelial cells). A clear lineage-dependent pattern has emerged: lymphoid subtypes show broad HSP-state accessibility, myeloid subtypes show selective HSP-state formation, and stromal subtypes show low accessibility with candidate HSP-high states mainly in Fibroblast-like cells and lymphatics. GO results consistently support a proteostasis-centered HSP core, including protein folding, response to unfolded or misfolded proteins, heat response, chaperone activity, and protein refolding.

Several limitations remain. The datasets are derived from different published atlases, so communication results should be interpreted only within each cell type or subtype rather

than as direct cross-lineage interactions. HSP-high cluster assignment still includes manual review of DotPlot, heatmap, and HSP score patterns; spatial niche explanations are currently mechanistic hypotheses rather than proven causes. Further spatial, clinical, and experimental validation will be required before final causal conclusions can be made.

II. Current Achievements

Please give a summary of your current achievement, e.g., research direction identified, technical reports/conference/journal papers produced or other academic activities attended.

The main achievement in this period is the construction of a structured cross-lineage HSP-state analysis framework and the completion of major result modules for human tumor single-cell data. In lymphoid cells, all analyzed B, CD4 T, CD8 T, and NK subtypes showed HSP-high clusters, supporting a broadly accessible immune stress-associated state. In myeloid cells, HSP-high clusters were detected in selected macrophage subtypes (FN1, IL1B, INHBA, LYVE1, NLRP3, and SPP1) and most dendritic-cell subtypes (AS DC, cDC1, cDC2, DC_RUNX3, DC_TRIM33, LAMP3+ DC, LC-like, and pDC), while C1QC/GPNMB/ISG15 macrophages and DC_GTF2IRD1/DC_MKI67 did not show separable HSP states. In stromal cells, most fibroblast and endothelial subtypes showed dispersed HSP expression rather than separable HSP clusters, with candidate HSP-high states identified in Fibroblast-like cells and lymphatics.

The analysis produced subtype-level HSP visualization, state separation metrics, DEG and GO summaries, trajectory evidence, CellChat summaries, and alternative stress-program scores. The results indicate that HSP-high cells share an upregulated proteostasis program, while downregulated functions differ by lineage: immune activation, adhesion, cytokine, phagocytosis, and communication-related terms are reduced in immune lineages, whereas translation, oxidative phosphorylation, ATP metabolism, and mitochondrial respiration are reduced in stromal HSP-high states. These outputs now provide a coherent technical basis for a manuscript outline and for presentation figures.

In addition, one related co-first-author paper was published during this reporting period: "A Pan-cancer Microenvironment Atlas Constructed by DECEPTICONx" in iMetaMed (published on 2026-01-17). This publication and the current HSP-state project together

strengthen my research direction in AI-assisted single-cell analysis, tumor microenvironment modeling, and data-driven therapeutic hypothesis generation.

III. Work Plan for Next 6 Months

Please specify in details.

No.	Content	Completion Date
1	Finalize the integrated HSP-state technical report, audit all subtype-level evidence, and prepare a manuscript-ready result framework.	2026-08
2	Add validation analyses using independent datasets, clinical annotations, spatial transcriptomics or tissue localization evidence where available.	2026-10
3	Prepare publication figures, manuscript draft, and downstream AIDD hypotheses linking HSP-state biology to therapeutic.	2026-12

IV. Lists

Papers Submitted/Revised/Accepted

(Past 6 months)

- [1] Yida Gu #, Fulan Deng #, Lianchong Gao, Zhiyu Chen, Jianping Chen, Jiale Wu, Zengqiang Xia, Zhijian Li, Fangkaikai Cheng, Jie Chen, Henry H. Y. Tong, Wantao Chen, Xiaoqing Pan, Xin Zou, Jie Hao. “A Pan-cancer Microenvironment Atlases Constructed by DECEPTICONx”, *iMetaMed*, Journal, **Published**, 2026-01-17.

(Accumulated)

- [2] Fulan Deng, Jiawei Zou, Miao Chen Wang, Yida Gu, Jiale Wu, Lianchong Gao, Yuan Ji, Henry Tong, Jie Chen, Wantao Chen, Lianjiang Tan, Yaoqing Chu, Xin Zou, Jie Hao. “DECEPTICON: a correlation-based strategy for RNA-seq deconvolution inspired by a variation of the Anna Karenina principle”, *Briefings in Bioinformatics*, B Journal, **Published**, 2025-05-03.
- [3] Lianchong Gao, Yujun Liu, Jiawei Zou, Fulan Deng, Zheqi Liu, Zhen Zhang, Xinran Zhao, Lei Chen, Henry Tong, Yuan Ji, Huangying Le, Xin Zou, Jie Hao. “Deep scSTAR: leveraging deep learning for the extraction and enhancement of phenotype-associated features from single-cell RNA sequencing and spatial transcriptomics data”, *Briefings in Bioinformatics*, B Journal, **Published**, 2025-05-01.

Events Attended

V. Technical Report

☐ None ☐ Attached

Signature of
Student



(Fulan Deng)

Date 2026-05-29